

Ayaan Asif

Software Engineer | Systems, Simulation and Scientific Computing

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<https://ayaan000.github.io/Simulator/> <https://ayaan000.github.io/ThirdDegreeBurn>

Education

University of Toronto

Sept 2022 – Jun 2026

Hon. BSc. Computer Science Specialist • Astrophysics Minor

Relevant Coursework: Operating Systems Design and Implementation, Algorithm Design Analysis & Complexity, Parallel Programming, Databases, Data Visualization, Computer Graphics, Physics-Based Animation, Cryptography

Experience

AI Trainer — Data Annotation Tech

Toronto, ON • Oct 2023 – Present

- Evaluate large-scale AI models across multiple technical domains by designing fine-grained rubrics focused on correctness, reasoning quality, safety, and consistency.
- Worked across diverse technical domains including systems programming, cloud infrastructure, and full-stack development to validate model behavior.

Game Developer — Level 9

Kolkata, India • May 2023 – Jun 2023

- Developed a 3D endless runner in which dinosaurs chase the player who dodges procedurally generated obstacles.
- Implemented rigid-body kinematics, collision handling, and gameplay logic in Unity (C#). Worked independently in coordination with the client.

Game Developer — Mayabious Art LLP

Kolkata, India • Aug 2021 – Sep 2021

- Created a 2D platformer. Developed character controllers, animation graphs, and interaction systems in Unity/C#.
- Created, rigged, and integrated Blender assets into Unity environments.

Some Projects

Type Ia Supernova Photometry Pipeline • Python, NumPy, Astropy, Photutils

- Built an end-to-end FITS-processing workflow using Fast Fourier Transforms, aperture photometry, and calibration.
- Measured light-curve evolution and developed intuition for astronomical signal processing.
- Validated light-curve extraction accuracy through calibration and noise analysis on astronomical datasets.

Fluid & Motion Simulations • C++, Python, Unity

- Implemented physics-based animation systems using meshes, constraint solvers, and numerical methods.
- Explored coordinate descent, reduced-order models, rigid and affine transformations, and fluid dynamics.
- Built multiple simulation prototypes to study numerical stability and performance trade-offs.

Scientific Hardware & Spectrometer Prototyping • CAD, Optics, 3D Printing

- Designed and fabricated optical mounts and assemblies using Onshape and Creality systems.
- Built early phone-based spectrometer prototypes; performed initial spectral extraction tests.

AstraMail — Email Migration Pipeline • Python, SQL, Regex

- Created data-cleaning and reconstruction pipelines for inbox deduplication, threading, and migration.

Game Development Collection (Demo on website) • Unity, C#

- Dino Runner (3D): Procedural obstacles, raptor AI, physics-based movement.
- NyanKnives: Parallax scroller, where Nyan Cat dodges knives while flying through space.
- QuickMath: Timed mental-math quiz engine.

Skills

Languages: Python, C++, C, C#, Java, JavaScript, SQL, Bash, R, Assembly, HTML/CSS

Systems & Tools: Linux, Multithreading, Memory Models, Git, Docker, React, Node.js, CUDA

Scientific & Graphics: NumPy, SciPy, PyTorch, Astropy, Photutils, Unity, Blender

Hardware & CAD: Onshape, Cura, G-code, 3D Printing, Spectroscopy

Spoken Languages: Native English and Hindi; Conversational French, Spanish and Bengali